

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 13 · Integration

Questions

Student book · questions with writing space

- 63 pages
- Past papers up to June 2026

Integration

Your classified route

33

questions

249

marks

4

families

10

variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Integration

Families and variants

F01 • Algebraic preprocessing and indefinite integration

V01 Direct sum of positive, negative and fractional powers

V02 Expand a product then integrate

V03 Expand or divide a radical or rational quotient

F02 • Recovering a curve from its first derivative

V01 Integrate f' and use one point

V02 Find parameter from gradient or line, then recover the curve

V03 Recover the curve and construct a tangent or normal

F03 • Recovering from second-derivative information

V01 Integrate twice using gradient and point conditions

V02 Use a second-derivative condition to find a parameter

F04 • Multi-constraint integration synthesis

V01 Determine multiple coefficients before recovering f

V02 Recovery plus a transformation or extra derivative demand

Algebraic preprocessing and indefinite integration

WHAT THIS FAMILY ASKS

Expand, divide or rewrite an integrand as a sum of powers, integrate term by term and include the arbitrary constant.

VARIANT ROUTE

V01 Direct sum of positive, negative and fractional powers

V02 Expand a product then integrate

V03 Expand or divide a radical or rational quotient

14 Complete Classified questions

68 total marks

1. Find, in simplest form,

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

(4)

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1. Find

$$\int 12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4} dx$$

giving each term in simplest form.

(5)

DR ESAM AHMED

1. Find

$$\int \left(\frac{8x^3}{5} - \frac{2}{3x^4} - 1 \right) dx$$

giving each term in simplest form.

(4)

DR ESAM AHMED

1. Find

$$\int \left(10x^5 + 6x^3 - \frac{3}{x^2} \right) dx$$

giving your answer in simplest form.

(4)

DR ESAM AHMED

1. Find

$$\int \left(10x^4 - \frac{3}{2x^3} - 7 \right) dx$$

giving each term in simplest form.

(3)

DR ESLAM AHMED

1. Find

$$\int \left(8x^3 - 6\sqrt{x} - \frac{2}{5x^3} \right) dx$$

giving your answer in simplest form.

(4)

DR ESAM AHMED

1.

$$y = 4x^3 + \frac{3}{2\sqrt{x}} - 8 \quad x > 0$$

Find, in simplest form,

(a) $\frac{dy}{dx}$

(3)

DR ESLAM AHMED

1.

$$y = 4x^3 + \frac{3}{2\sqrt{x}} - 8 \quad x > 0$$

Find, in simplest form,

(b) $\int y \, dx$

(3)

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1. Find

$$\int (2x - 5)(3x + 2)(2x + 5)dx$$

writing your answer in simplest form.

(5)

DR ESLAM AHMED

2. Find

$$\int (x-3)^2(2x+5) dx$$

writing the answer in simplest form.

(5)

DR ESLAM AHMED

1. Find

$$\int \left(3x + \frac{2}{x} \right)^2 dx$$

giving the answer in simplest form.

(5)

DR ESAM AHMED

4. Find

$$\int \frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} dx$$

writing each term in simplest form.

(6)

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2. Given that

$$(x - 5)(2x + 1)(x + 3) \equiv ax^3 + bx^2 - 32x - 15$$

where a and b are constants,

(a) find the value of a and the value of b .

(2)

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2. Given that

$$(x - 5)(2x + 1)(x + 3) \equiv ax^3 + bx^2 - 32x - 15$$

where a and b are constants,

(b) Hence find

$$\int \frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} dx$$

writing each term in simplest form.

(5)

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3. Find

$$\int \frac{4x^5 + 3}{2x^2} dx$$

giving your answer in simplest form.

(5)

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3. Find

$$\int \frac{(x+3)^2}{3\sqrt{x}} dx$$

writing your answer in simplest form.

(5)

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Recovering a curve from its first derivative

WHAT THIS FAMILY ASKS

Integrate f' , determine the constant from a point and connect the derivative to tangent or normal information when present.

VARIANT ROUTE

V02 Find parameter from gradient or line, then recover the curve

V03 Recover the curve and construct a tangent or normal

11 Complete Classified questions

107 total marks

8. A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

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A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

$$k = -5$$

(b) Hence find $f(x)$.

(4)

DR ESAM AHMED

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

The point $P(4, 12)$ lies on the curve.

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(ii) find the value of c

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant

$$k = \frac{1}{4}$$

(b) Hence find the value of $f''(x)$ at P .

(3)

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The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant

$$k = \frac{1}{4}$$

(c) Find $f(x)$.

(4)

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In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(2)

(2)

8.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(c) Find $f(x)$, writing your answer in simplest form.

(4)

DR ESAM AHMED

In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

A point P lies on C .

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$

(a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P ,

(6)

8.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve C has equation $y = f(x)$, $x > 0$

A point P lies on C .

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

Given that the x coordinate of P is $\sqrt{3}$

(b) find an equation of the normal to C at P .

(2)

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8.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A curve C has equation $y = f(x)$, $x > 0$

A point P lies on C .

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

- (c) Find $f(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

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8. A curve C has equation $y = f(x)$, $x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C .

Given that

- $f'(x) = 10 + \frac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

- (a) (i) find the value of k
- (ii) find the equation of the normal to C at P .
Give the answer in the form $y = mx + c$, where m and c are constants.

(6)

8. A curve C has equation $y = f(x)$, $x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C .

Given that

- $f'(x) = 10 + \frac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

(b) Find $f(x)$.

(5)

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6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$

- $f'(x) = \frac{32}{3x^2} + 3 - 2(\sqrt[3]{x})$

(a) find the equation of the tangent to C at P . Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

DR ESAM AHMED

6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$

- $f'(x) = \frac{32}{3x^2} + 3 - 2(\sqrt[3]{x})$

(b) Find, in simplest form, $f(x)$.

(5)

DR ESAM AHMED

6. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

(a) (i) find the value of the gradient at P

(ii) Hence find the equation of the tangent to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

DR ESAM AHMED

6. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

(b) Find $f(x)$, simplifying your answer.

(7)

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5.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$
- the point $P(9, 8)$ lies on C

(a) find, in simplest form, $f(x)$

(5)

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In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

Given that

- $f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$

- the point $P(9, 8)$ lies on C

The line l is the normal to C at P

(b) Find the coordinates of the point at which l crosses the y -axis.

(4)

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

(b) find $f(x)$ giving the coefficients in simplest form.

(5)

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7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers to be found.

(4)

DR ESAM AHMED

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \frac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(b) Find $f(x)$.

(6)

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10. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$

- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

DR ESAM AHMED

10. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = 6x - \frac{(2x-1)(3x+2)}{2\sqrt{x}}$
- the point $P(4, 12)$ lies on C

(b) find $f(x)$, giving each term in simplest form.

(6)

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Recovering from second-derivative information

WHAT THIS FAMILY ASKS

Use f'' either to reconstruct f through two integrations or to determine a parameter in f' before integrating once.

VARIANT ROUTE

V01 Integrate twice using gradient and point conditions

V02 Use a second-derivative condition to find a parameter

6 Complete Classified questions

53 total marks

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

- (a) the equation of the tangent to the curve at P , writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

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11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(b) $f(x)$.

(8)

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11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \quad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P ,

- (a) find the equation of the normal at P , writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

DR ESAM AHMED

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \quad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P ,

(b) find $f(x)$.

(8)

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11. A curve C has equation $y = f(x)$, $x > 0$

Given that

- $f''(x) = 4x + \frac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

DR ESAM AHMED

11. A curve C has equation $y = f(x)$, $x > 0$

Given that

- $f''(x) = 4x + \frac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(b) find $f(x)$, writing your answer in simplest form.

(6)

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10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a .

(3)

DR ESAM AHMED

10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(b) Hence find $f(x)$.

(4)

DR ESLAM AHMED

7. The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

(a) find the value of A .

(4)

DR ESAM AHMED

7. The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

(b) find $f(x)$, giving each term in simplest form.

(5)

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10.

In this question you must show all stages of your working.

The curve C has equation $y = f(x)$, $x > 0$

Given that

- the point $P(2, 8\sqrt{2})$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k ,

(4)

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In this question you must show all stages of your working.

Given that

- the point $P(2, 8\sqrt{2})$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(b) find $f(x)$, giving your answer in simplest form.

(4)

Multi-constraint integration synthesis

WHAT THIS FAMILY ASKS

Resolve several independent constants or combine integration with an additional transformation or derivative demand.

VARIANT ROUTE

V01 Determine multiple coefficients before recovering f

2 Complete Classified questions

21 total marks

9. (i) Find

$$\int \frac{(3x + 2)^2}{4\sqrt{x}} dx \quad x > 0$$

giving your answer in simplest form.

(5)

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(ii) A curve C has equation $y = f(x)$.

Given

- $f'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $f(x)$.

(6)

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In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24$$

(2)

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(b) find, in simplest form, $f(x)$

(7)

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \frac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

Given that point P is transformed to the point Q ,

(1)